Recovering reaction kinetics and homeostatic regulation from metabolite dynamics with graph neural networks

Abstract

A metabolic network is a bipartite graph in which metabolites are coupled through reactions, each with its own rate constant and stoichiometry. We pose the *inverse* problem: given only the time series of metabolite concentrations and the bipartite graph structure, recover the components that generated the dynamics — the per-reaction rate constants k_j, the substrate-to-rate kinetic law, and the per-metabolite homeostatic regulation. We replace the unknown functions of the forward model with learnable MLPs that operate on the metabolite–reaction graph and a vector of learnable rate constants, and train the graph neural network (GNN) end-to-end to predict dc/dt. On synthetic mass-action systems with known ground truth we study rate-constant recovery and the identifiability obstacles that arise from scale ambiguity, functional compensation, and homeostasis absorption. We outline the path from synthetic data to real metabolomics — starting from fully parameterised kinetic models from BioModels (yeast glycolysis), constraint-based genome-scale models (yeast-GEM), and high-frequency real-time metabolomics during nutrient transitions. This is a companion methods note to our connectome-dynamics studies, which apply the same sign-/structure-locked GNN inverse-modelling recipe to neural circuits.

1 Introduction

Cellular metabolism is a large, sparse reaction network: hundreds of metabolites are interconverted by hundreds of enzyme-catalysed reactions, each proceeding at a rate set by an intrinsic rate constant and the instantaneous concentrations of its substrates. The *forward* problem — predicting concentration dynamics given the network, the kinetics and the rate constants — is classical biochemistry. The *inverse* problem is harder and is our subject: given measured concentration trajectories and the graph of which metabolites enter which reactions, recover the rate constants and the functional form of the kinetics. This is the metabolic analogue of the connectome-to-dynamics inverse problems we study elsewhere, and it shares their central difficulty: the mapping from parameters to dynamics is nonlinear, high-dimensional, and degenerate — many parameter combinations produce nearly identical trajectories.

2 Metabolic networks as bipartite graphs

A metabolic network is naturally a **bipartite graph** with two node types: metabolites and reactions. Edges connect only nodes of different type, and each edge carries an integer *stoichiometric coefficient*. A single-partite metabolite–metabolite graph cannot represent the system, because a reaction couples *several* substrates and products at once through a shared rate constant: collapsing it to pairwise metabolite edges would lose the information that those metabolites participate in the same reaction.

Stoichiometric matrix. The connectivity is summarised by the stoichiometric matrix $\mathbf{S} \in \mathbb{R}^{n\times m}$ (n metabolites, m reactions), where S_{ij} is the signed coefficient with which metabolite i enters reaction j: negative for substrates (consumed), positive for products (produced). $\mathbf{S}$ is sparse (each reaction touches only a handful of metabolites), integer-valued (typically in $\{-2, -1, 0, +1, +2\}$), and for mass-balanced reactions its columns sum to zero.

3 Forward model: reaction kinetics and homeostasis

Mass-action kinetics. Under the law of mass action a reaction proceeds faster when its substrates are more abundant, with each substrate raised to the power of its stoichiometric coefficient. For reaction j with rate constant k_j,

$$v_j \;=\; k_j \prod_{k\in \mathrm{sub}(j)} c_k^{|S_{kj}|}, \tag{1}$$

so a reaction consuming two molecules of a substrate depends quadratically on its concentration, and one consuming a single molecule depends linearly. The product over substrates (*multiplicative* aggregation) encodes the requirement that *all* substrates be present simultaneously, and the resulting nonlinear coupling is what allows autocatalytic cycles to sustain oscillations. A simpler *additive* aggregation $v_j = k_j \sum_k c_k^{|S_{kj}|}$ treats substrates as contributing independently and is a useful surrogate for enzyme-mediated reactions that are not strictly mass-action.

Full dynamics. Combining reaction fluxes through $\mathbf{S}$ with a per-metabolite homeostatic restoring force gives the complete forward model

$$\frac{dc_i}{dt} = \underbrace{-\lambda_i\,(c_i - c_i^{\mathrm{baseline}})}_{\text{homeostasis}} + \underbrace{\sum_{j=1}^{m} S_{ij}\, v_j}_{\text{reaction dynamics}}\quad , \tag{2}$$

with v_j from Eq. (1). The homeostatic term pulls each metabolite toward a baseline level c_i^{baseline} at a metabolite-specific rate λ_i; crucially it is *small* relative to the reaction fluxes, which is what makes it hard to recover (Sect. 5). Eq. (2) spans a family of regimes used for synthetic data: pure reaction ($\lambda = 0$, additive), homeostatic ($\lambda > 0$, additive, convergent), and oscillatory (autocatalytic 3-cycles, multiplicative, sustained oscillations).

4 Inverse problem: the GNN parameterisation

Given observed trajectories $\{c_i(t)\}$ and the bipartite graph, we recover the generating model by replacing every unknown function in Eqs. (1) and (2) with a learnable component on the graph:

$$\frac{dc_i}{dt} = \underbrace{\mathrm{MLP}_{\mathrm{node}}(c_i, a_i)}_{\approx -\lambda_i (c_i - c_i^{\mathrm{baseline}})} + \sum_{j=1}^{m} S_{ij}\, k_j \underbrace{\prod_{k\in \mathrm{sub}(j)} \mathrm{MLP}_{\mathrm{sub}}\big(c_k, |S_{kj}|\big)}_{\approx k_j\, c_k^{|S_{kj}|}}\,. \tag{3}$$

The GNN operates on the metabolite–reaction graph by message passing: substrate states are mapped through $\mathrm{MLP}_{\mathrm{sub}}$, aggregated per reaction (multiplicatively for mass-action), scaled by the learnable rate constant k_j, and distributed back to metabolites through $\mathbf{S}$; the node update $\mathrm{MLP}_{\mathrm{node}}$ adds the self-regulation term. Training minimises the prediction error on dc/dt end-to-end. The learnable parameters are listed in Tab. 1.

Parameter	Type	Role
$a_i \in \mathbb{R}^d$	embedding vector	per-metabolite identity
k_j	scalar (m of them)	per-reaction rate constant (learned in log-space)
$\mathrm{MLP}_{\mathrm{sub}}(c_k, \lvert S_{kj}\rvert)$	MLP	discovers the kinetic law $c_k^{\lvert S_{kj}\rvert}$
$\mathrm{MLP}_{\mathrm{node}}(c_i, a_i)$	MLP	discovers homeostasis $-\lambda_i(c_i - c_i^{\mathrm{baseline}})$

Table 1: Learnable components of the GNN inverse model (Eq. (3)). The stoichiometric matrix $\mathbf{S}$ is either frozen from ground truth ("S given" mode, the focus here) or made learnable ("S learning" mode).

Function discovery. The GNN is not told the kinetic law. $\mathrm{MLP}_{\mathrm{sub}}$ receives a substrate concentration and its stoichiometric coefficient and must learn the power-law response $c_k^{|S_{kj}|}$ from data alone — in particular it must distinguish linear (c^1) from quadratic (c^2) dependence as the coefficient changes. Likewise $\mathrm{MLP}_{\mathrm{node}}$ must recover the subtle, small-magnitude homeostatic signal without absorbing dynamics that belong to the reaction terms.

5 Identifiability

The three learnable components interact and can compensate for one another, producing degenerate solutions with high prediction accuracy but poor parameter recovery:

- **Scale ambiguity.** The product $k_j\,\mathrm{MLP}_{\mathrm{sub}}$ is invariant under $k \to \alpha k$, $\mathrm{MLP}_{\mathrm{sub}} \to \mathrm{MLP}_{\mathrm{sub}}/\alpha$. Without anchoring, the MLP absorbs a global scale and shifts every k_j. A centring regulariser pins mean$(\log k)$ to the known range centre.
- **Functional compensation.** If $\mathrm{MLP}_{\mathrm{sub}}$ settles on a wrong exponent (*e.g.* $c^{1.5}$ instead of c^2), the rate constants can partly compensate by re-scaling, masking the error.
- **Homeostasis absorption.** $\mathrm{MLP}_{\mathrm{node}}$ can grow and soak up dynamics that should be explained by reactions, biasing the recovered rate constants.

These are the metabolic counterparts of the sign/scale degeneracies that motivate the structure-locking and regularisation priors in our connectome-dynamics models.

6 Toy model

The toy model is a synthetic rank-50 oscillatory mass-action network (100 metabolites, 256 reactions, stoichiometry $\mathbf{S}$ given, no homeostasis), generated with known rate constants and fit leak-free (the loss reads only the concentration dynamics, never a ground-truth label). The *rate-constant recovery* R^2 (learned vs. true $\log_{10} k_j$) is 0.75 ± 0.04 across four seeds (Tab. 2); for the published seed it is raw 0.74 over all 256 reactions, or trimmed 0.98 after excluding the 4.3% (11/256) outliers, at slope 1.00

(Fig. 1c). The two learned functions match ground truth (Fig. 1a,b): MLP_{sub} recovers the substrate law $c^{|s|}$ and MLP_{node} (homeostasis) stays ≈ 0, the correct value for this regime. Rollout quality is measured by the *Pearson* correlation between the learned free autoregressive rollout and the true trajectory: it is high early (0.96 up to time 25) and degrades as integration error accumulates (0.85 at time 50, 0.74 up to the divergence onset at time $\approx$124), after which the rollout diverges (Fig. 1d). The rate constants are recovered, but single-step training places no constraint on multi-step stability, so the autocatalytic dynamics eventually blow up. Rate-constant recovery and rollout are different questions: this model recovers the parameters but is not a stable simulator.

We then asked whether a recurrent training curriculum closes the stability gap: the rollout horizon is grown across epochs (single-step, then $100, 200, \ldots, 1000$ integration steps, with a co-ramped learning rate and gradient clipping). A naive run is a cautionary tale about the metric. The free trajectory no longer diverges (Fig. 2d), and the *pooled* rollout Pearson even rises ($0.74 \to 0.83$) — but the pooled correlation is dominated by *between-metabolite* level differences and merely rewards staying bounded. Measured honestly *per metabolite* (each rollout against its own true trajectory), fidelity *collapses* from 0.47 (single-step) to 0.18: the curriculum trades the single-step model's correct-but-unstable oscillations for a smooth, bounded, dynamically dead trajectory that no longer reproduces the peaks. Rate-constant recovery collapses in step (Fig. 2c): the correlation between learned and true $\log_{10} k_j$ falls from 0.87 to 0.07 (raw $R^2 = 0.004$, 90% outliers), and the homeostasis MLP_{node}, correctly ≈ 0 under single-step training, drifts to a spurious nonzero function. The multi-step objective admits a smooth *degenerate* solution — wrong rate constants and a spurious homeostasis term reproduce a damped trajectory without the mechanism — the same right-dynamics/wrong-parameters degeneracy seen on glycolysis. The lesson is twofold: report *per-metabolite* rollout fidelity (the pooled number is flattering), and recognise that the naive curriculum does not buy stability so much as suppress dynamics. Whether *any* curriculum achieves a genuinely good rollout (high per-metabolite fidelity *and* stability *and* preserved k-recovery) is an open question we are addressing with a sweep of schemes — horizon caps, learning-rate decay, k-anchoring (freezing the recovered rate constants past the single-step warmup), soft tail losses, and longer warmups; results pending.

Seed	training completed	rate-constant recovery R^2	rollout Pearson
77 (published winner)	100%	0.742	0.74
42	100%	0.705	0.77
79	90%	0.808	0.37
7	90%	0.738	0.72
mean ± s.d. (4 mature seeds)		$\mathbf{0.748 \pm 0.037}$	0.65 ± 0.16
123	30% (stopped)	0.722 (rising)	0.65

Table 2: Cross-seed reproduction of the leak-free k_j-recovery working point. Rate-constant recovery R^2 clusters tightly at 0.75 ± 0.04 over four seeds; seed 123 was stopped early at 0.72 and still rising. Rollout Pearson is the global correlation between the learned free rollout and the true trajectory over the convergent window (before divergence); it is far more variable (0.65 ± 0.16) and never stable — every seed recovers the rate constants but none yields a stable simulator.

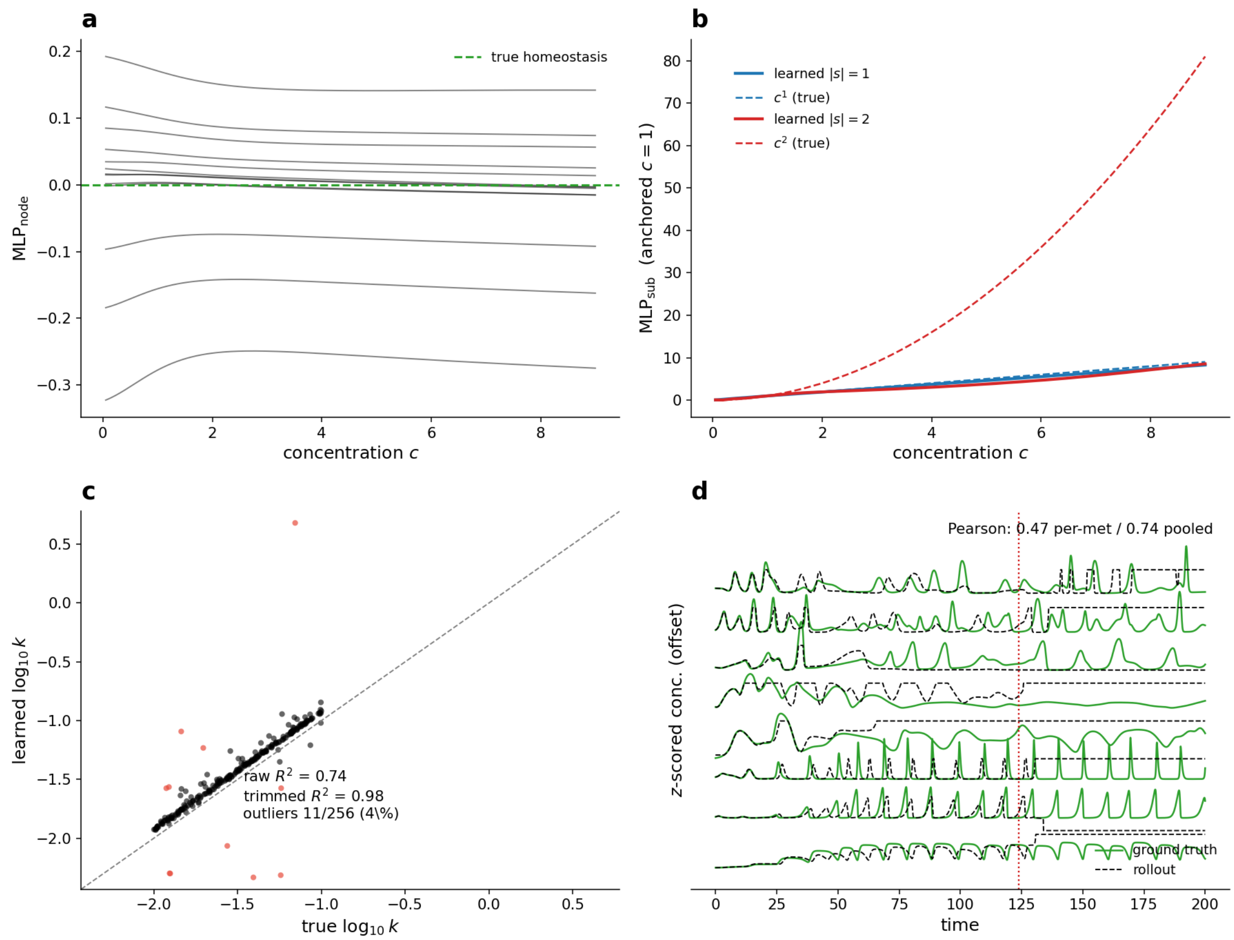

Figure 1: **Toy-model results dashboard (leak-free, seed 77).** **(a)** MLP$_{\text{node}}$ per metabolite (homeostasis); it stays ≈ 0 (dashed: the true zero homeostasis of this regime). **(b)** MLP$_{\text{sub}}(c, |s|)$ (solid) vs. the true power law $c^{|s|}$ (dashed) for $|s|{=}1, 2$, anchored at $c{=}1$: the substrate law is recovered for $|s|{=}1$ (the regime the toy contains), while the $|s|{=}2$ curve is an out-of-support extrapolation. **(c)** Rate-constant recovery: learned vs. true $\log_{10} k_j$ for all 256 reactions after the MLP$_{\text{sub}}$ scalar correction (dashed: identity, slope 1.00); 11 red points (4.3%) exceed the 0.3-dex outlier gate, raw $R^2{=}0.74$, trimmed $R^2{=}0.98$. **(d)** Free rollout of the eight highest-variance metabolites, z-scored and stacked: ground truth (green) vs. the learned autoregressive rollout (dashed black). It tracks the oscillations early and drifts until it diverges at time $\approx$124 (red line); convergent-window rollout Pearson 0.47 per metabolite (0.74 pooled). Rate constants are recovered, but the single-step-trained model is not a stable simulator.

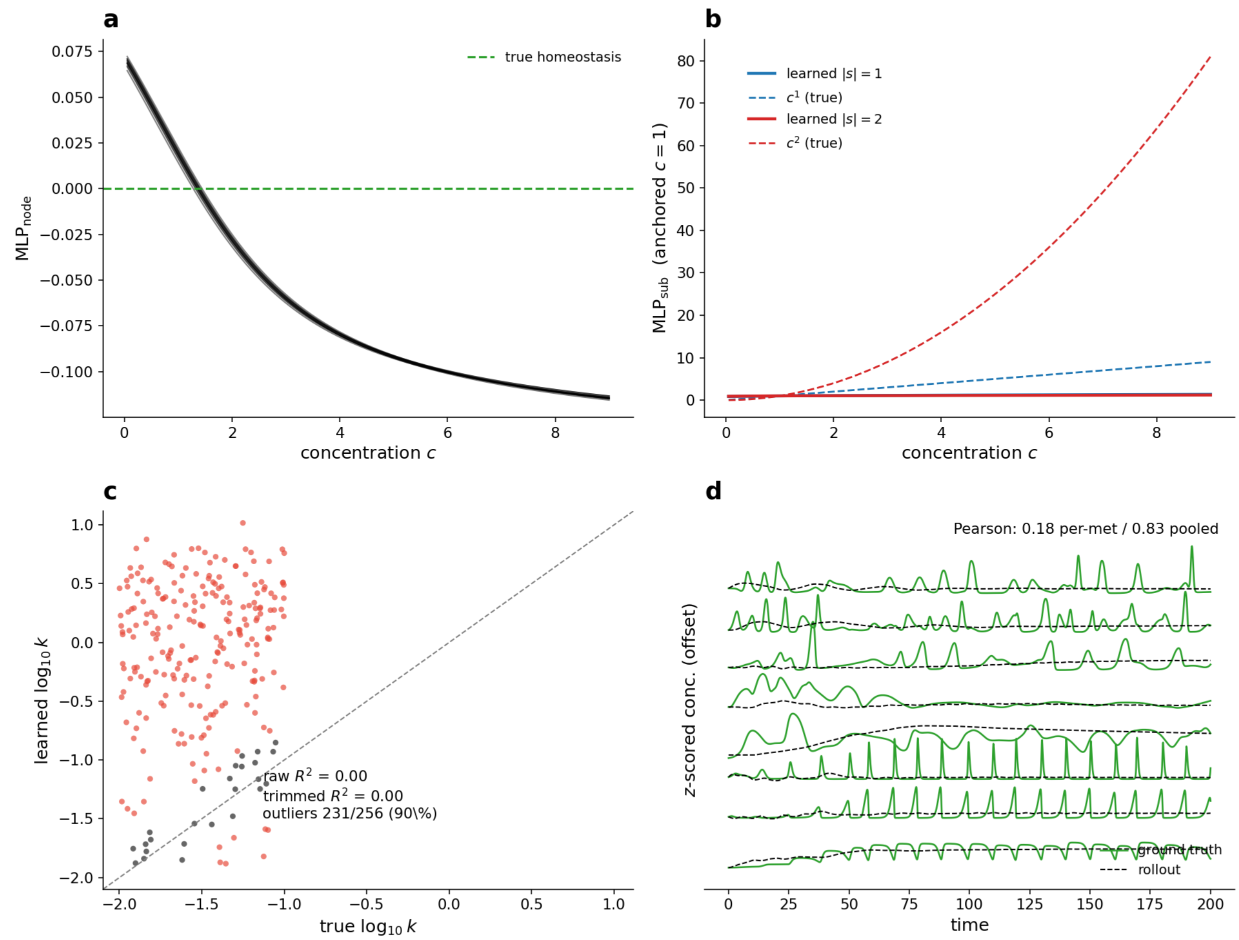

Figure 2: **Naive recurrent curriculum: stable but dynamically dead.** Same dashboard as Fig. 1, for a model trained with the horizon-growing curriculum (single-step → 1000 steps). **(d)** The free rollout no longer diverges, and the *pooled* Pearson rises to 0.83 — but per metabolite (each rollout vs. its own true trajectory) fidelity *drops* to 0.18 (from 0.47 single-step): the predicted traces are smooth and bounded but no longer reproduce the oscillations. The pooled number is flattering because it is dominated by between-metabolite level differences. **(c)** Rate-constant recovery has *collapsed*: a structureless cloud (90% outliers, raw R^2=0.00, correlation with truth 0.07 vs. 0.87 single-step). **(a)** And $\mathrm{MLP}_{\mathrm{node}}$, correctly ≈ 0 under single-step training, has drifted to a spurious nonzero homeostasis. The multi-step objective is satisfied by a smooth *degenerate* solution — wrong k plus spurious homeostasis — that suppresses dynamics rather than learning them. (**(b)** $|s|$=2 is out-of-support extrapolation here, as in Fig. 1.)

A different lever closes the stability gap where the curriculum could not: supplying a known *external stimulus*. The toy is autonomous, and a long autonomous rollout of a linear oscillatory system collapses to a damped solution under any multi-step objective (above). Feeding a known time-varying drive to a subset of metabolites at every rollout step — the boundary-input convention used by the calcium models — changes the picture. With the drive supplied and ten distinct driven trajectories for training, even *single-step* training yields a rollout that is stable over the full window (no divergence) while preserving rate-constant recovery ($R^2 = 0.78$), and it stays bounded on a held-out trajectory with a different drive (Fig. 3). The external input anchors the dynamics the way it does in the connectome models. Per-metabolite fidelity (0.33) does not yet exceed the stimulus-free early-window value (0.47), but the stimulus model holds that fidelity across the *entire* trajectory rather than only up to divergence; whether the curriculum, now anchored by the drive, lifts it further without the degeneracy was then tested directly, by running the stimulus-driven *curriculum* (three horizon schedules, each to a fixed 12 h compute budget; Fig. 4). The answer is negative: the curriculum does not improve on single-step training. Among the three schedules the best k-recovery comes from the *capped* horizon (cap-100): the rollout stays stable and dynamically alive over the full window (per-metabolite Pearson 0.31), so the stimulus does prevent the dead-collapse that the stimulus-free curriculum suffered (per-metabolite ≤ 0.23, k destroyed). But the rollout fidelity (0.31) merely matches single-step training (0.33, Fig. 3) rather than exceeding it, *and* rate-constant recovery degrades — from $R^2 = 0.78$ / 4% outliers (single-step) to $R^2 = 0.44$ / 14% outliers, now failing the $\leq 10\%$ gate. The uncapped and zebrafish-style schedules are worse still on k (32% and 84% outliers). The earlier warm-up reading of a high k-recovery held only at the short horizon; once training actually reaches a multi-step horizon the curriculum erodes k, exactly the stability/identifiability tension of Fig. 2, merely softened (not removed) by the drive. **Single-step training with the external stimulus remains the best toy configuration.**

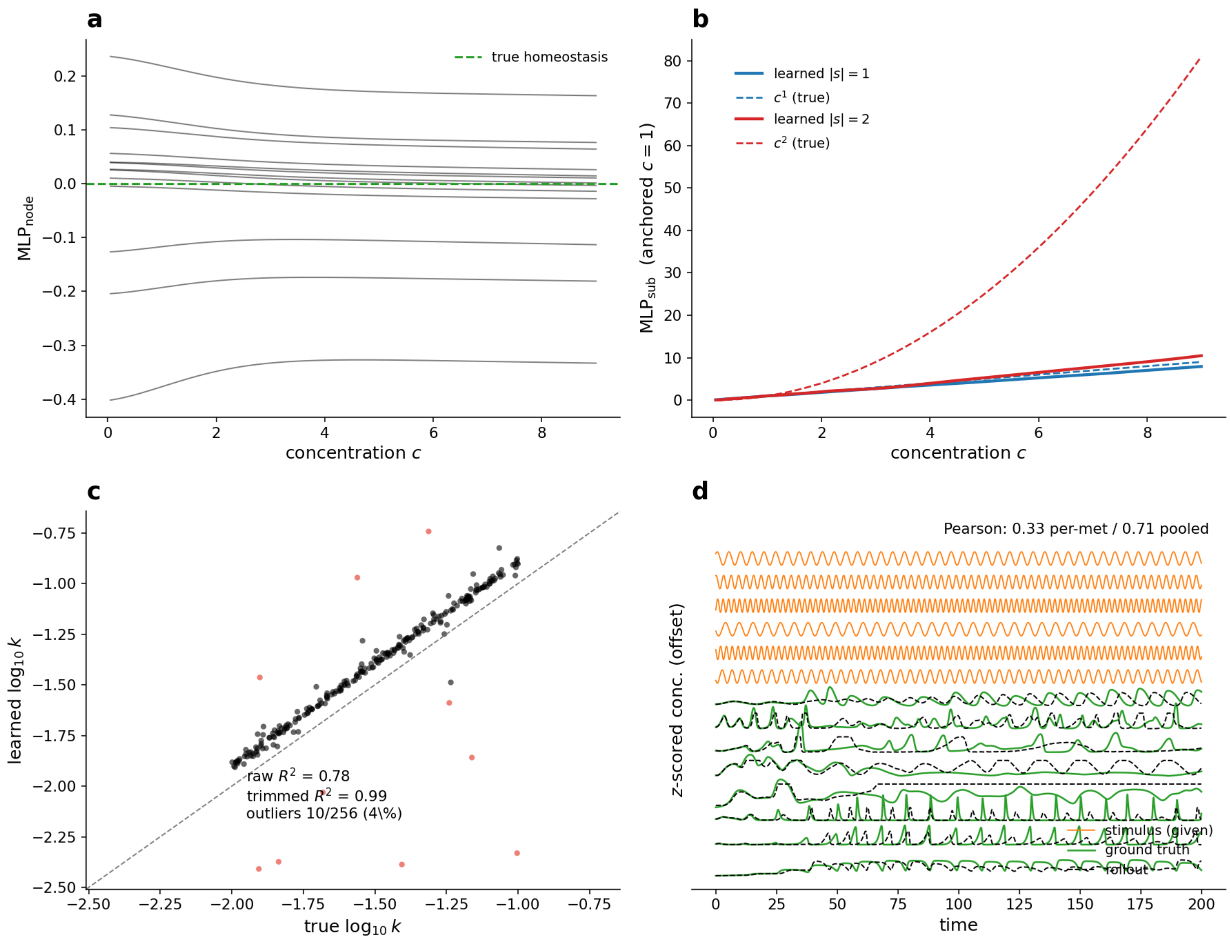

Figure 3: **An external stimulus anchors the rollout.** Same dashboard, for single-step training on ten trajectories driven by a known time-varying stimulus (orange in **d**, the given input, plotted as in the glycolysis figure). **(d)** The free rollout (black) is now stable over the full window with no divergence (contrast Fig. 1d, which diverges at $t \approx 124$, and Fig. 2d's degeneracy); per-metabolite Pearson 0.33. **(c)** Rate-constant recovery is preserved ($R^2 = 0.78$, 4% outliers) — the stimulus stabilises the rollout *without* the parameter collapse the curriculum caused. On a held-out trajectory with a different drive the rollout stays bounded (per-metabolite 0.20).

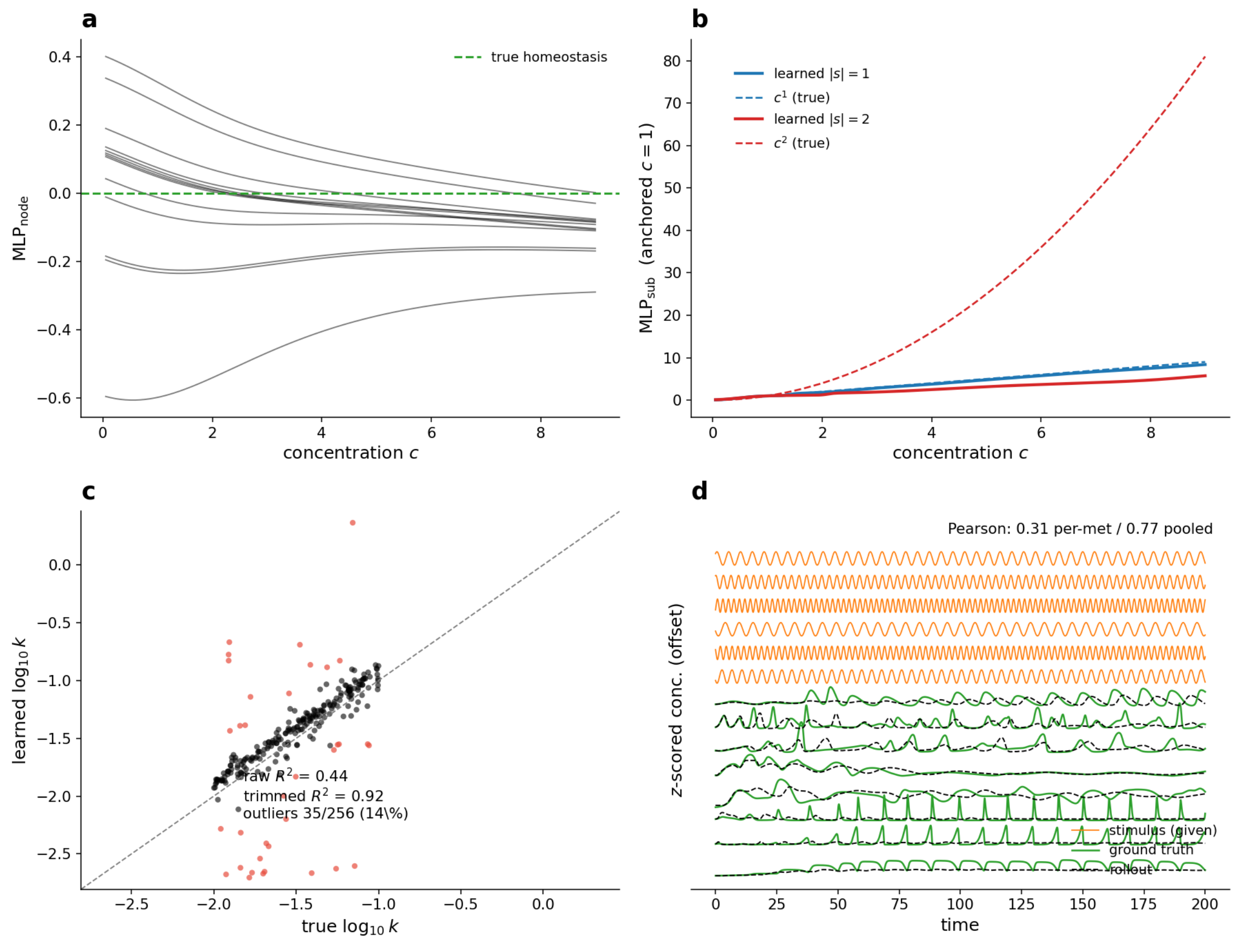

Figure 4: **The recurrent curriculum, even anchored by the stimulus, does not beat single-step training.** Best of three stimulus-driven curriculum schedules (the capped cap-100 horizon, trained to a 12 h budget). **(d)** The free rollout (black) under the given drive (orange) is stable over the full window with no divergence and tracks the ground-truth oscillations (green), but per-metabolite Pearson is only 0.31 — matching, not exceeding, single-step training (Fig. 3, 0.33). **(c)** Rate-constant recovery has degraded to $R^2 = 0.44$ (trimmed 0.92) with 14% outliers (red), above the $\leq 10\%$ gate and well below the single-step value (0.78, 4%); **(a)** MLP_{node} also drifts to a spurious nonzero function. The stimulus prevents the dead-collapse of the stimulus-free curriculum but the multi-step objective still trades away k identifiability for no rollout gain.

Intrinsic noise as an identifiability lever — and the right form for a conserved system. Stochastic forcing *inside* the dynamics (distinct from measurement noise on the observations) can diversify a trajectory and, in a degenerate problem, help break identifiability. But the *form* of the noise matters when the state obeys a conservation law. Adding an additive term $\sigma\,\xi_i(t)$ directly to the concentrations — the Euler step of the stochastic differential equation $dc = f(c)\,dt + \sigma\,dW$ — is stoichiometrically *unsound*: it injects or removes mass that no reaction produced, so we do not use it. The sound alternative perturbs the reaction *rates*: $k_j(t) = k_j\,(1 + \sigma\,\xi_j(t))$, a fluctuation in enzyme activity. Each reaction still conserves mass through S and only its flux varies, so every trajectory stays physical (sub-10% mass drift) while the data diversify (activity rank 35 $\rightarrow$ 50 at $\sigma = 0.3$). The rollout sweep (Fig. 5) shows the effect directly: as σ grows the simulated ground truth becomes visibly jagged while the deterministic model rolls out smoothly, recovering the underlying clean dynamics from noisy data at low noise. But for *recovery* the rate noise is *benign then harmful*: at $\sigma = 0.1$ it is unchanged from $\sigma = 0$ (trimmed $R^2 = 0.96$, 4.7% outliers; the raw $R^2 = 0.85$ sits inside the cross-seed band), then degrades monotonically ($\sigma = 0.3$: 21% outliers; $\sigma = 0.5$: 60%, as the noise corrupts the trajectory itself). So getting the noise physically right does not change the outcome: with S given, k is *already* identifiable at $\sigma = 0$ (Fig. 1), so there is no degeneracy left for any noise to break. The lever would matter only in a genuinely k-degenerate regime (e.g. S unknown), which we leave to future work.

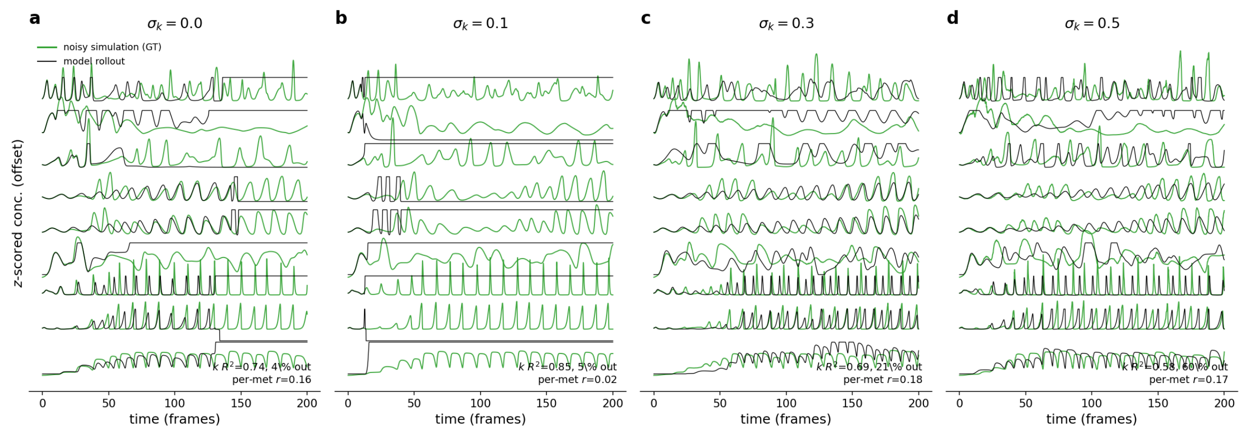

Figure 5: **Stoichiometrically-sound rate-noise sweep (rollout traces).** Intrinsic noise applied to the reaction *rates*, $k_j(1 + \sigma_k \xi)$ (enzyme-activity fluctuation; mass-conserving), at four levels. Each panel shows the noisy simulated ground truth (green) and the trained model's deterministic free rollout (black), z-scored and stacked, with the leak-resistant k-recovery R^2 and per-metabolite rollout Pearson annotated. As σ_k grows the simulation becomes visibly jagged while the model rollout stays smooth; k-recovery is unchanged at $\sigma_k = 0.1$ and then degrades (21% then 60% outliers), confirming that on this S-given toy the (sound) noise neither helps nor is needed to identify k.

7 Toward real metabolomics

The framework is developed and validated on synthetic data where ground truth is known; the goal is real metabolomic time series. Three concrete datasets, now vendored in the repository, stage that transition from fully parameterised kinetics, through genome-scale topology, to a real recording. None of the three is solved yet; this section describes what each provides and what stands between it and a result.

Rung 1 — yeast glycolysis kinetic model (BioModels the published yeast glycolysis kinetic model). A small, fully parameterised kinetic model of glycolysis in *S. cerevisiae* (Smallbone et al., 2013), imported from SBML. After separating dynamic from boundary species it gives 20 internal metabolites (+9 fixed boundary species), 30 reactions including iso-enzymes (three hexokinase, three pyruvate-decarboxylase, three glyceraldehyde-3-phosphate variants), with real stoichiometry $\mathbf{S}$ and measured initial concentrations. The stoichiometry is sparse (77 nonzero entries) and almost entirely unit-valued — coefficients are ± 1 (40 negative, 35 positive) with a single -2 and a single -3 — but, unlike the all-$|s|$=1 toy, it is genuinely *multi-substrate*: of the 30 reactions 15 are first-order in substrate, 13 are second-order ($|s|$=2), and one is fourth-order, so there is real kinetic curvature to recover. Crucially the rate laws are *not* mass-action but *Michaelis–Menten* (MM), the saturating enzyme-kinetic law $v = V_{\max} c/(K_m + c)$: the reaction velocity rises with substrate and plateaus, where $V_{\max}$ (the *maximal velocity*, set by enzyme amount and turnover) is the per-reaction rate constant to be recovered — the MM analogue of the mass-action k_j — and K_m (the *half-saturation constant*) is the substrate concentration at which $v = V_{\max}/2$. Glycolysis mixes reversible/irreversible MM, Hill, and allosteric forms, so $v_j = V_{\max,j} \prod_k \left[c_k/(K_{m,kj} + c_k)\right]^{|S_{kj}|}$ rather than a power law. This is the cleanest available ground truth: known $V_{\max}/K_m$ let us score parameter recovery directly. *Status:* the SBML import, **S** extraction, and a Michaelis–Menten simulator are in place and generate a 12,000-frame dataset, and we port the connectome-style *autoregressive curriculum* (per-epoch horizon ramp $10 \to 300$, soft tail-loss, LR co-ramp, gradient clipping) to stabilise the rollout. Two cautions, both found by being sceptical of an initially good number. First, the test rollout looked catastrophic ($R^2 = -390$) only because the evaluator omitted the model's *boundary stimulus*; supplying it, the curriculum model reproduces the trajectory well (global $R^2 = 0.99$, Fig. 6). Second, that success is misleading: it is *in-sample* (a single trajectory, no held-out run), the per-metabolite R^2 is only 0.15, and the leak-resistant test fails outright — the recovered V_{max} correlate with ground truth at $R^2 \approx 0$ (slope 0.06, 73% outliers, Fig. 8). The model therefore reproduces the dynamics with the *wrong* kinetics: a degenerate fit. A *direct held-out test* confirms it does not generalise: rolling the trained model out on an unseen trajectory (a different boundary stimulus, comparable rank, Fig. 7) collapses the fit from global Pearson 0.997 in-sample to 0.28 (global R^2 $0.99 \to 0.05$; per-metabolite R^2 strongly negative) — the predicted traces settle to the wrong levels and no longer track the true dynamics. The good in-sample rollout was effectively memorisation of the single training trajectory. The curriculum fixed rollout *stability*; the real open problem on real Michaelis–Menten data is parameter *identifiability* — the same scale/compensation degeneracy as the synthetic setup, but here unbroken. (One robust falsification survived both metrics: the config with the lowest *training* loss had the worst rollout *and* the worst recovery, so training loss is not a proxy for either.)

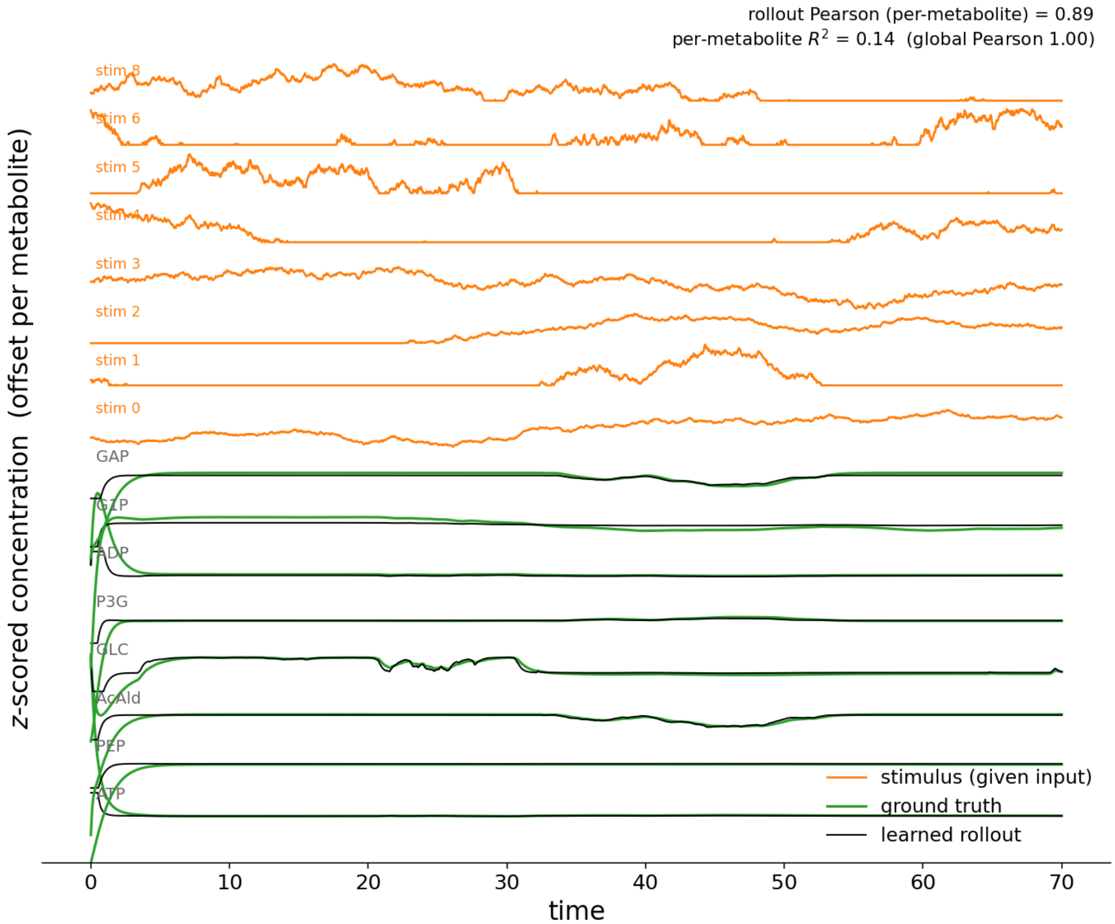

Figure 6: **Rung 1 glycolysis: free-running rollout, ground truth vs learned** (the curriculum model, stimulus supplied). The eight highest-variance metabolites, z-scored and stacked by a vertical offset: ground truth (green) vs. the learned free autoregressive rollout (dashed black); the nine *given* external stimulus channels (orange, top) are the known boundary drive supplied to the rollout, not predicted. The rollout tracks the trajectory with no divergence; rollout Pearson vs. the true trajectory is 0.997 globally (0.89 per-metabolite). But note the metabolites mostly relax to a steady state under the strong given drive — so the high Pearson is partly stimulus-driven settling, and per-metabolite R^2 is only 0.14 (the settled levels drift). This is *in-sample* (single trajectory, no held-out run), and despite the good-looking rollout the rate constants are not recovered ($V_{\max}$ recovery $R^2 \approx 0$, Fig. 8).

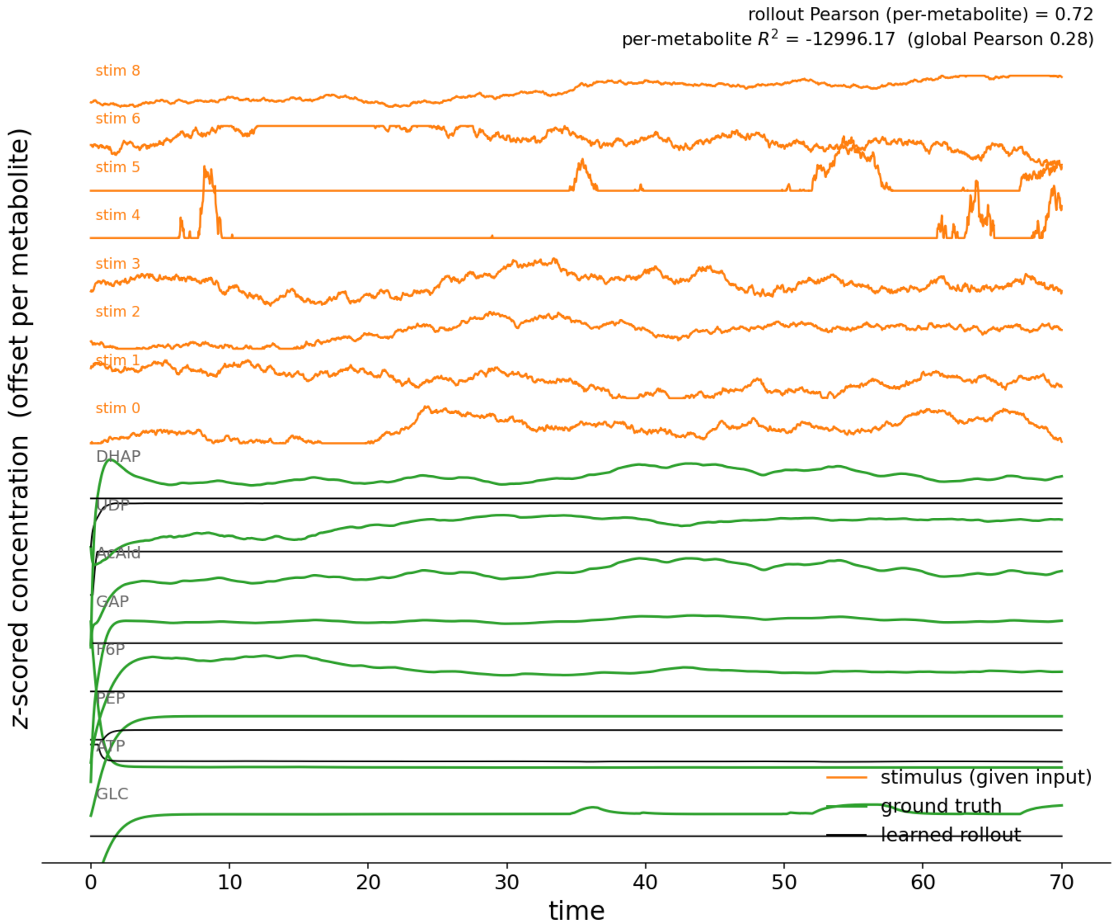

Figure 7: **Rung 1 glycolysis: the held-out generalisation test.** The same trained model rolled out on an *unseen* trajectory — a different given stimulus (orange, top) and a different initial state, comparable rank — never used in training. Unlike the in-sample case (Fig. 6), the learned rollout (dashed black) settles to the *wrong* levels and no longer tracks ground truth (green): global Pearson falls from 0.997 to 0.28 (global R^2 0.99 $\rightarrow$ 0.05; per-metabolite R^2 strongly negative). The in-sample fit was memorisation of one trajectory, not a learned mechanism — direct evidence for the degeneracy implied by the failed $V_{\max}$ recovery.

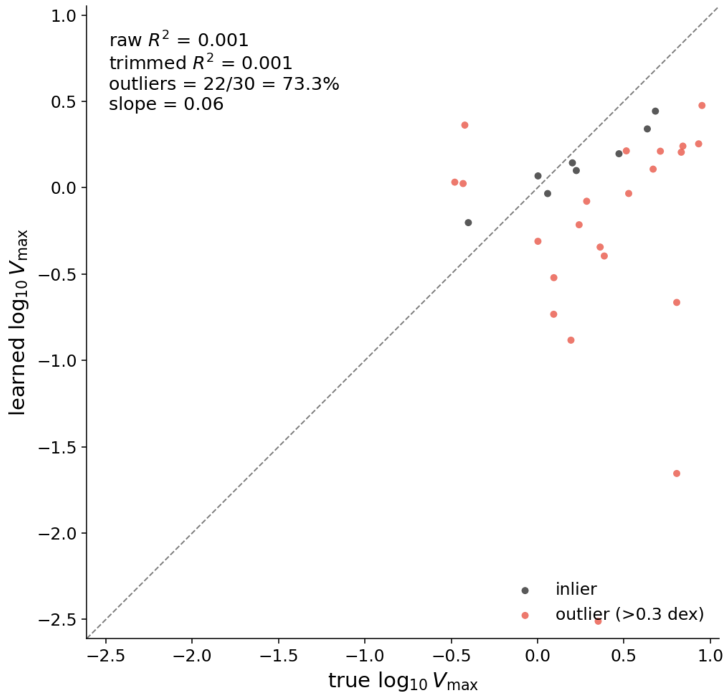

Figure 8: **Rung 1 glycolysis: V_{max} recovery fails** — the leak-resistant test. Learned vs true $\log_{10} V_{max}$ for the 30 reactions: $R^2 \approx 0$, slope 0.06, 22/30 (73%) outliers (fails the 10% gate). Despite the near-perfect in-sample rollout (Fig. 6), the model recovers essentially *none* of the true kinetics — a degenerate fit (right dynamics, wrong mechanism).

Does $\mathrm{MLP}_{\mathrm{sub}}$ learn the glycolysis kinetics, and does the log-exp form help? A natural suspect for the recovery failure is that $\mathrm{MLP}_{\mathrm{sub}}$ cannot represent the substrate kinetics. Unlike the synthetic power-law toy, glycolysis is Michaelis–Menten: the substrate law *saturates*, $[c/(K_m+c)]^{|s|}$, rather than growing as $c^{|s|}$. We compare the learned $\mathrm{MLP}_{\mathrm{sub}}$ against that MM truth for the plain MLP and the log-space parameterisation $g = \exp(\mathrm{MLP}(\log c, |s|))$ (Fig. 9). Representation is *not* the bottleneck for the well-populated orders: $|s|$=1 (15 reactions) is tracked by both forms, and $|s|$=2 (13 reactions) is partly captured (plain) or mildly undershot (log-space). Only $|s|$=4 — a *single* reaction — is unconstrained, where the plain MLP extrapolates wildly and log-space flattens: with one example there is no signal to fit it (the same scarcity-of-high-order-reactions limit discussed below). Crucially, the log-exp form does *not* rescue Rung 1: $V_{\max}$ recovery is $R^2 = 0.00$ (plain) and 0.04 (log-space), both failing outright. The exponential ansatz, which added curvature on the power-law toy, is mismatched to a *saturating* law; and in any case the limiting factor is not how $\mathrm{MLP}_{\mathrm{sub}}$ is parameterised but the Michaelis–Menten identifiability degeneracy ($V_{\max}$ and K_m trade off) — the substrate function can be represented, yet the rate constants are not recovered.

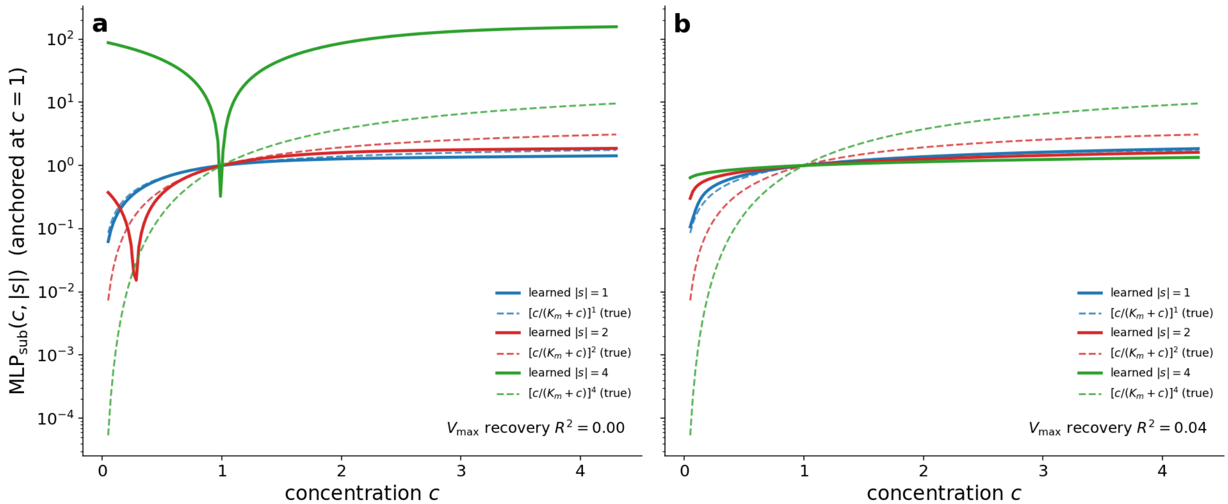

Figure 9: **Rung 1: does** $\mathrm{MLP_{sub}}$ **learn the glycolysis substrate kinetics?** Learned $\mathrm{MLP_{sub}}(c, |s|)$ (solid) vs the true Michaelis–Menten law $[c/(K_m+c)]^{|s|}$ (dashed; saturating, not a power law) for the glycolysis substrate orders $|s|=1, 2, 4$, log y-axis, anchored at $c=1$. **(a)** plain MLP, **(b)** log-space MLP. The well-populated $|s|=1$ (and largely $|s|=2$) shapes are captured; $|s|=4$ is a single reaction and unconstrained (plain blows up, log-space flattens). Either way $V_{\max}$ recovery fails ($R^2 = 0.00$ plain / 0.04 log-space), so the substrate-function form is not the limiting factor.

The limit is signal, not capacity. Two controls on a synthetic $|s|\geq 2$ regime (where the substrate orders, unlike glycolysis's one $|s|=4$ reaction, are populated enough to test) localise the residual failure. (i) Training on four diverse trajectories instead of one barely helps (outliers 37.5%→32%), so it is not a data-volume / identifiability-by-replication problem. (ii) A *structured* substrate function $g = c^{a(|s|)}$ — which can represent any exponent *exactly* — still fails (outliers 30.9%): its learned exponent map saturates at $a(s)=1.0, 1.45, 1.64, 1.71, 1.73, 1.75$ for $s=1\ldots6$, against the true $1, 2, 3, 4, 5, 6$. It nails $a(1)$ (499 reactions) but never reaches the high exponents — because those reactions are *rare* (15/12/8/2 reactions at $|s|=3/4/5/6$). The unrecoverable ∼31% are exactly the $|s|\geq 2$ reactions, across all three parameterisations. So the bottleneck for high-order kinetics is neither MLP capacity nor data volume but the *scarcity of high-order reactions*: too little signal to identify a steep exponent. The fix is a structural prior (e.g. integer exponents) or a regime richer in high-$|s|$ reactions — not a better generic function approximator.

Rung 2 — consensus genome-scale model (yeast-GEM). The community-curated genome-scale model of *S. cerevisiae* (Lu et al., 2019): 2,806 metabolites, 4,131 reactions across 14 compartments. It supplies a large, real stoichiometric matrix $\mathbf{S}$ and flux bounds but *no* kinetic rate laws — the true inverse problem. The intended use is to extract a central-carbon subgraph (glycolysis + TCA + fermentation, ∼100–150 reactions), drive it with dynamic-FBA or literature kinetics to generate in-silico dynamics, and fit the GNN in "S given" mode on a realistic topology. *Status:* not started beyond vendoring the model; subgraph extraction and a dynamics generator are still to be written.

Rung 3 — real-time metabolomics during nutrient transitions (real data). High-frequency flow-injection mass spectrometry of live microbial culture (Link et al., 2015), now downloaded as the paper's supplementary tables. For *E. coli* the released data give a metabolite×time intensity matrix of 247 annotated ions over 119 evenly spaced time points in three biological replicates (a long batch course), plus a dedicated starvation→growth switch series of 302 ions over ∼148 points; companion *B. subtilis* and *Y. lipolytica* sets are included. Each ion carries a KEGG annotation (444 distinct KEGG IDs for *E. coli*), though many ions are mass-ambiguous (one m/z maps to several candidate metabolites) and values are *relative* ion intensities, not absolute concentrations. *Status:* the data are in hand but not yet ingested; the work is to deduplicate/annotate ions, build $\mathbf{S}$ for the measured subset from an *E. coli* genome-scale model (*e.g.* iJO1366) via KEGG→BiGG mapping, and rescale the relative intensities. There is *no* kinetic ground truth here, so success can only be judged by held-out prediction, not by recovered k_j. A parallel mammalian direction is chronic jugular microdialysis in freely moving mice (Nardin et al., 2025) (∼123 metabolites every 7.5 min over ∼8 h; 10 components explain 73% of variance).

What transfers and what changes. The core recipe transfers: the bipartite graph, message passing over reactions, function discovery via $\mathrm{MLP}_{\mathrm{sub}}/\mathrm{MLP}_{\mathrm{node}}$, and log-space rate-constant recovery. Real data adds, all at once, the obstacles the synthetic setup omits: non-mass-action (saturating, allosteric) kinetics that $\mathrm{MLP}_{\mathrm{sub}}$ must learn; a partially known and incomplete $\mathbf{S}$ (only measured metabolites, ambiguous identities); two orders of magnitude fewer, unevenly spaced time points ($\sim 10^2$ vs $\sim 10^3$); relative rather than absolute concentrations; and external drivers (the nutrient switch) that must enter as a term $f_{\mathrm{ext}}(t)$ in Eq. (2).

What we expect. We do *not* expect the real-data rungs to work soon. The realistic near-term target is Rung 1: get the GNN to recover the published $V_{\max}/K_m$ of a real kinetic model under Michaelis–Menten kinetics — a contained problem with ground truth, but one the current code does not yet solve. Rung 2 is a topology-realism check with no kinetic ground truth. Rung 3 (real *E. coli*) is the genuine goal but also where every difficulty above appears together; a first "result" there means only predicting held-out frames of measured dynamics, not recovering mechanism. Tab. 3 summarises the gap. The honest reading: one validated synthetic working point, three real rungs at the data-and-scaffolding stage, and substantial method development (saturating kinetics, honest homeostasis, incomplete-network handling, relative-concentration rescaling) still required before any real-data claim.

Identifiability on a real network topology (imposed kinetics). Short of full real data, we can already probe the identifiability question on *real* metabolic topology by imposing synthetic Michaelis–Menten kinetics (random $V_{\max}/K_m$, hence known ground truth) on the real stoichiometry $\mathbf{S}$ of two networks: *E. coli* core (72 metabolites, 71 enzymatic reactions after dropping biomass/exchange pseudo-reactions) and a cytosolic central-carbon subgraph of yeast-GEM (208 metabolites, 120 reactions). With $\mathbf{S}$ given the model reproduces the dynamics — pooled rollout Pearson 0.99, per-metabolite 0.55 (*E. coli*) and 0.58 (yeast) — yet recovers *none* of the kinetics: $V_{\max}$ recovery is $R^2 \approx 0$ with 100% of reactions outside the 0.3-dex gate, on *both* networks (Fig. 10). This is the glycolysis degeneracy in its most extreme form, and the cause is dynamical, not structural: these closed networks relax to a steady state within a few frames, so their trajectories have activity rank ≈2 — far below the synthetic toy (rank 50, oscillatory, where k *is* recovered at $R^2 = 0.74$) or even glycolysis (rank 5, 73% outliers). Identifiability therefore tracks the *information content of the dynamics*, not the size of the network or the method: a closed metabolic subnetwork at steady state is information-starved. Recovering kinetics on real topology will require dynamics that hold the system *off* its fixed point — a sustained external drive, or the nutrient switch of Rung 3 — to raise the rank.

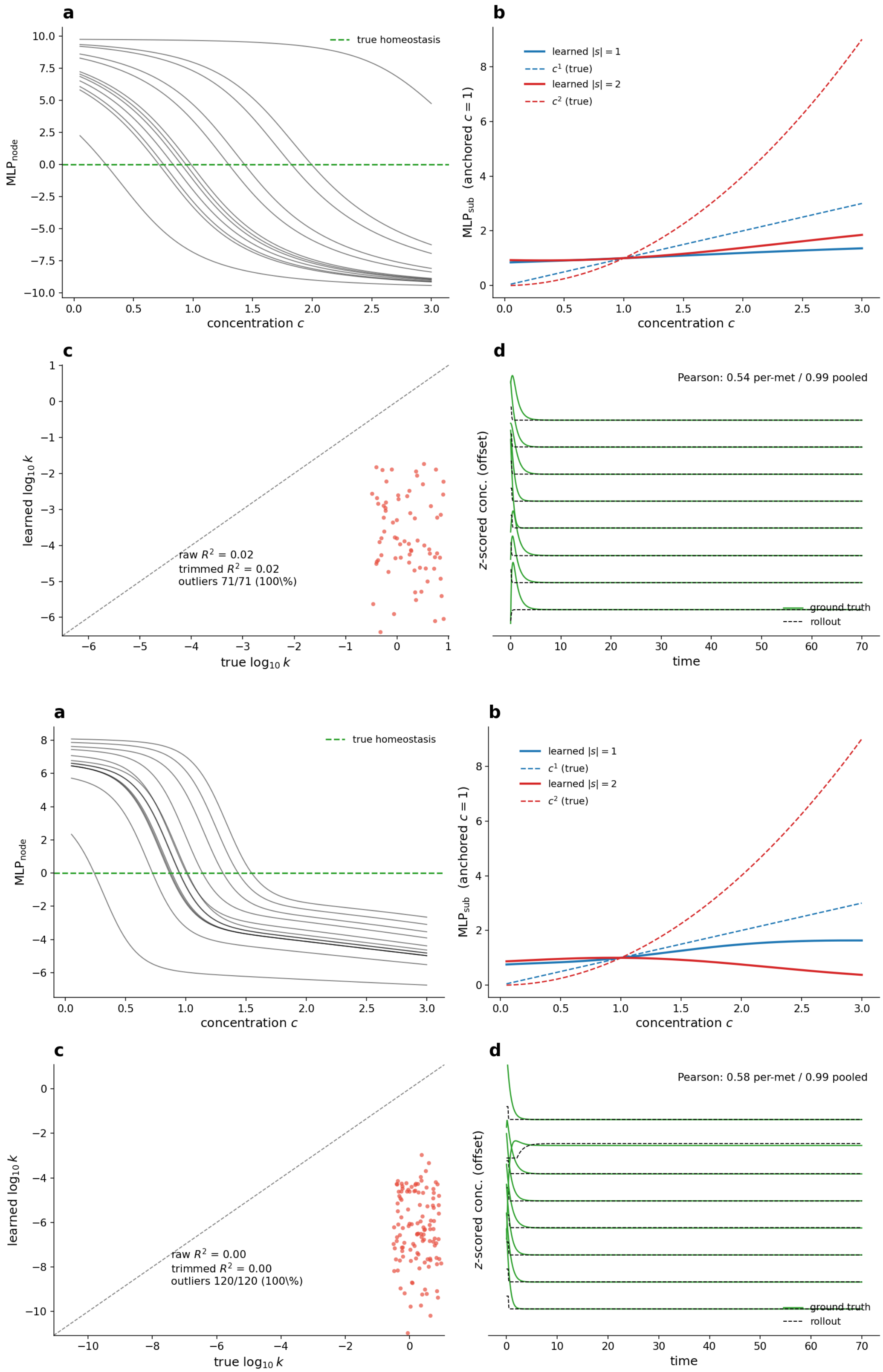

Figure 10: **Identifiability on real topology with imposed MM kinetics.** Top: *E. coli* core (72×71); bottom: yeast-GEM cytosolic subgraph (208×120). Real stoichiometry, synthetic Michaelis–Menten kinetics (so $V_{\max}/K_m$ are known ground truth), **S** given. Each dashboard: **(a)** $\mathrm{MLP}_{\mathrm{node}}$, **(b)** $\mathrm{MLP}_{\mathrm{sub}}$, **(c)** $V_{\max}$ recovery, **(d)** free rollout. The rollout is plausible (pooled Pearson 0.99, per-metabolite 0.55–0.58) but $V_{\max}$ recovery is total failure ($R^2 \approx 0$, 100% outliers) — the closed networks settle to steady state (activity rank ≈2), carrying too little information to constrain the kinetics.

Raising identifiability: drive the network off steady state. The failure above is a *data* problem, and it has a precise linear-algebra reading. With $\mathbf{S}$ given, recovering the kinetics is *linear in the parameters*, $\mathbf{A}\,\boldsymbol{\theta} = \mathbf{b}$, where each row is a measured dc/dt and the column multiplying a reaction's rate constant is built from its substrate concentrations over time. If a metabolite never varies (a network at steady state), its columns are constant or zero — null and sloppy directions in $\mathbf{A}$ — and those parameters are unidentifiable. Equivalently, linearising the dynamics as $\dot{c} = Jc + B\,u(t)$, an *autonomous* trajectory (u=0) collapses into the few slowest eigenmodes (activity rank ≈2), whereas an external source u on m metabolites drives the trajectory into the controllability subspace $\mathrm{span}\{B, JB, J^2B, \dots\}$, whose dimension grows with m and with the spectral richness of u. We therefore inject a known multi-frequency metabolite source (an additive open-system term $u_i(t)$, the analogue of the glycolysis boundary stimulus) on a subset of metabolites and sweep its breadth. The activity rank rises *linearly* with the number of driven metabolites — rank ≈m — on both real topologies (Fig. 11): from the rank-≈2 closed steady state to rank 19–21 at m=20 and ≈30 at m=40, clearing the rank-20 target. Information content, not network size, is the lever; the next step trains on these high-rank, stimulus-driven trajectories.

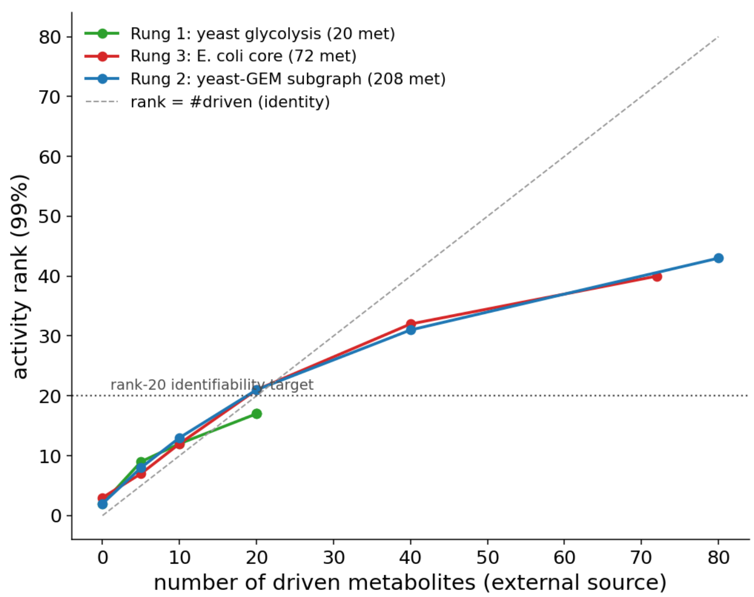

Figure 11: **An external metabolite source raises the activity rank ≈ linearly.** Activity rank (99%) of the simulated trajectory vs the number of metabolites carrying a known multi-frequency source drive, for all three rungs under imposed MM kinetics: Rung 1 glycolysis (20 met), Rung 3 *E. coli* core (72 met), Rung 2 yeast-GEM subgraph (208 met). Closed (0 driven) networks sit at rank ≈2; driving m metabolites gives rank ≈m (dashed identity) on all three, crossing the rank-20 identifiability target at m≈20. The drive is the data-side prerequisite for recovering kinetics.

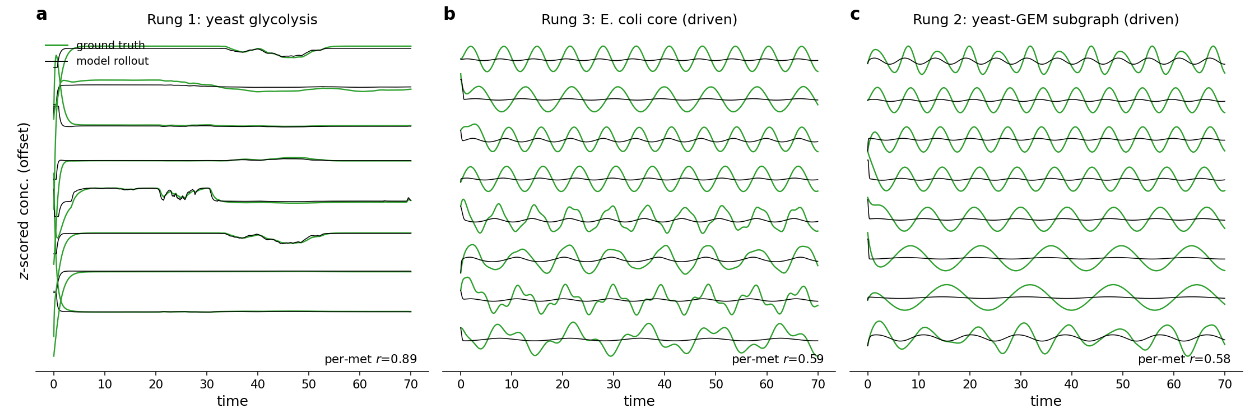

Figure 12: **Free-rollout traces under the external drive, one panel per rung.** Eight highest-variance metabolites, z-scored and stacked: ground truth (green) vs the trained model's free rollout (black). **(a)** Rung 1 glycolysis (settles to steady state); **(b)** Rung 3 *E. coli* core and **(c)** Rung 2 yeast-GEM subgraph, both with the m=40 metabolite source (rank 38): the drive holds the networks in sustained, information-rich oscillation rather than a fixed point. The rollout tracks the driven dynamics (per-metabolite Pearson 0.58–0.59); recovering the *kinetics* from them is the open step (cf. Fig. 11).

Identifiability is not the bottleneck — learnability is. The GNN trained on the driven (rank-38) *E. coli* data recovers *none* of the kinetics ($V_{\max}$ $R^2 \approx 0$, 100% outliers), exactly as on the undriven net. Yet the inverse problem is *well-posed*. Given $\mathbf{S}$ *and the rate-law shape*, dc/dt is linear in the per-reaction scale, $A\,\theta = b$ with columns $A_{\cdot j} = \mathbf{S}_{\cdot j}\,\phi_j(c)$ ($\phi_j = \prod c^s$ for mass action; $\phi_j = \prod[c/(K_m{+}c)]^s$ with the true K_m for MM). Built exactly from the simulator's Jacobian (reconstruction residual 10^{-4}), this A is *full column rank* and well conditioned at *both* rank-2 (cond. 121) and rank-38 (cond. 31); a plain least-squares solve then recovers $V_{\max}$ *perfectly* ($R^2 = 1.00$) *and* that $V_{\max}$ free-rolls-out at per-metabolite Pearson 1.00, at both ranks. So the data always suffice — identifiability is not rank-limited — and a single correct solution nails parameters *and* rollout; the GNN simply does not find it.

The catch is that this R^2=1 is conditional on *knowing* ϕ. The GNN does not: it must learn the shape ($\mathrm{MLP}_{\mathrm{sub}}$) *and* the scale ($V_{\max}$) jointly, and that factorisation is where MM fails. A saturating shape with a free per-reaction K_m can absorb the per-reaction scale, so $V_{\max}$ drifts; the rigid power law of mass action cannot, so its scale is pinned and recovers — which is why the GNN recovers the mass-action toy (R^2=0.74) even though *its* design matrix is *worse* posed (rank-deficient, least-squares R^2=0.87). GNN success is thus decoupled from identifiability and set by shape×scale learnability. The lever is therefore the model, not the data: solve $V_{\max}$ by least squares *given the GNN's learned shape*, or constrain $\mathrm{MLP}_{\mathrm{sub}}$'s shape/scale — not more rank, drive, or trajectories.

Target	Network / kinetics	Ground truth	Status
Synthetic k_j (S given)	random / mass-action	k_j known	**works**, $R^2 \approx 0.87$
Synthetic homeostasis	random / mass-action	λ_i known	open (single-step blind)
Rung 1: yeast glycolysis	real / MM+Hill	$V_{\max}, K_m$ known	rollout in-sample; $V_{\max}$ $\mathrm{R}^2 \approx 0$ (degener
Real topology (*E. coli* core)	real S / imposed MM	$V_{\max}$ known	$V_{\max}$ $\mathrm{R}^2 \approx 0$, 100% out (rank-2 degener
Real topology (yeast-GEM subgraph)	real S / imposed MM	$V_{\max}$ known	$V_{\max}$ $\mathrm{R}^2 \approx 0$, 100% out (rank-2 degener
Rung 3: real *E. coli* data	inferred / unknown	none	data in hand; not ingested

Table 3: Where each target stands. Only the first row is a validated, leak-free result; the rest is the work ahead.

7.1 First look: are the four real datasets amenable?

Before training, we run the same cheap test on *all four* datasets measured by Link et al. (2015) — *E. coli*, *B. subtilis*, *Y. lipolytica*, and a hybrid sample — asking whether each recording carries the structure the GNN needs. The identical four-probe analysis is applied to every organism: SVD low-rank, per-ion lag-1 autocorrelation, example traces, and KEGG coverage against a common reference network (iJO1366; a proxy for core-metabolite mappability — exact for *E. coli*, approximate for the others). For the *Y. lipolytica* fit target the proper yeast network yeast-GEM raises coverage from 63% to 77% (127/164 ions), confirming the under-estimate from the bacterial proxy. Tab. 4 summarises and Fig. 13 shows the per-organism panels.

What the three probes compute, and how to read them.

- **SVD low-rank.** Stack the recording into a matrix $C \in \mathbb{R}^{N \times T}$ (N ions, T time points) and z-score each row over time, so the probe sees temporal *shape* not absolute scale. Its singular values $\sigma_1 \geq \sigma_2 \geq \ldots$ give the variance carried by each independent mode, $\sigma_i^2 / \sum_j \sigma_j^2$; rank_{99} is the smallest number of modes whose cumulative variance reaches 99%. *Interpretation:* it counts the effective dimensionality of the dynamics. A small rank_{99} relative to N means the metabolites move in a coordinated, low-dimensional way — shared drivers and strong correlations — which is exactly the structure a compact network model can fit; $\mathrm{rank}_{99} \approx N$ would mean every ion moves independently and no parsimonious mechanism exists. We compare the *fraction* rank_{99}/N against the synthetic regime ($\sim$45%).
- **Per-ion lag-1 autocorrelation.** For each ion's mean-subtracted trace $x(t)$, $\rho_1 = \sum_t x(t)\, x(t-1) \big/ \sum_t x(t)^2$ — the correlation of the signal with itself shifted by one frame; we report the median over ions and the fraction with $\rho_1 > 0.5$. *Interpretation:* it measures temporal smoothness. $\rho_1 \to 1$ means consecutive samples are nearly equal — a smooth, slowly varying trace whose finite-difference derivative dc/dt is well defined and carries signal; $\rho_1 \to 0$ means white-noise-like, so the measured dc/dt is dominated by noise. This is the decisive test for our setting, since the GNN learns $dc/dt = f(c)$: a derivative made of noise cannot be fit, however low-rank or well-annotated the data are.
- **KEGG coverage.** Each ion is annotated with one or more KEGG compound identifiers from its m/z (an ion matching several formulae is *ambiguous*); we intersect those identifiers with the metabolites present in a reference genome-scale network (iJO1366), and coverage is the number of ions hitting at least one model metabolite. *Interpretation:* it asks whether a stoichiometry $\mathbf{S}$ — the reaction graph the GNN operates on — can be built for the measured species. A metabolite

absent from every known reaction has no edges and cannot be modelled mechanistically, so high coverage means the measured set sits inside a real network; the *unambiguous* count (single KEGG per ion) flags how many identities are clean versus mass-degenerate.

Organism	ions×t	rank_{99}	autocorr med.	in iJO1366	dc/dt
E. coli	247×119	102 (41%)	0.27	190 (77%)	noisy
B. subtilis	239×119	98 (41%)	0.41	160 (67%)	noisy
Y. lipolytica	164×59	51 (31%)	**0.62**	103 (63%)	**smooth**
Hybrid	338×60	55 (16%)	0.24	155 (46%)	noisy

Table 4: Amenability of the four measured datasets. rank_{99} is the number (and fraction) of singular components for 99% variance; autocorr is the median per-ion lag-1 autocorrelation; coverage is the count mapped onto iJO1366. All four are low-rank and reasonably network-mappable; only *Y. lipolytica* has smooth (dc/dt-learnable) dynamics.

Does it generalise? Two of the three signals hold across all four datasets: the dynamics are *low-rank* ($\mathrm{rank}_{99} = 16$–41% of ions) and a large share of metabolites are core/conserved (coverage 46–77%). The third — temporal smoothness of dc/dt — does *not* generalise: *E. coli*, *B. subtilis* and the hybrid are noise-dominated (median lag-1 autocorrelation 0.24–0.41), whereas ***Y. lipolytica* is smooth** (median 0.62, 59% of ions above 0.5). The actionable consequence: the most promising target for an actual real-data fit is *not E. coli* (best network coverage but noisiest) but ***Y. lipolytica*** — smooth, learnable dynamics that are still low-rank and network-mappable. The hybrid is least amenable (lowest coverage, noisy), though most strongly low-rank. A real fit should therefore start from *Y. lipolytica*, restricted to its smooth, high-variance, network-mapped ions and judged by held-out prediction — not by mechanism recovery across the whole noisy measurement.

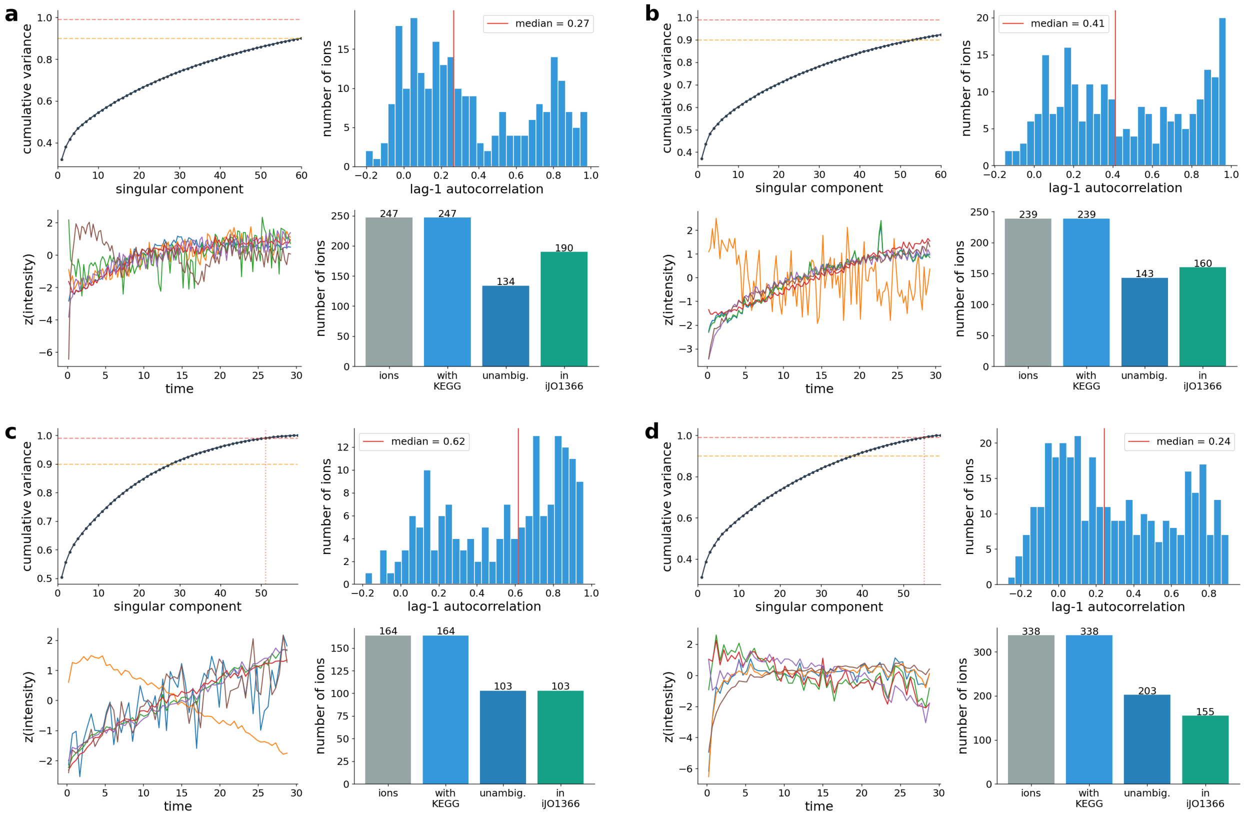

Figure 13: **Identical amenability probes across the four measured datasets: (a)** *E. coli*, **(b)** *B. subtilis*, **(c)** *Y. lipolytica*, **(d)** hybrid. Within each panel, top-left: SVD cumulative variance (low-rank); top-right: per-ion lag-1 autocorrelation; bottom-left: six highest-variance traces (z-scored); bottom-right: KEGG coverage against iJO1366. Low-rank structure and network coverage recur in all four; only *Y. lipolytica* shows smooth, slowly varying traces and a right-shifted autocorrelation distribution.

References

Smallbone, K., Messiha, H.L., Carroll, K.M., *et al.* (2013). A model of yeast glycolysis based on a consistent kinetic characterisation of all its enzymes. *FEBS Letters* 587(17), 2832–2841. BioModels the published yeast glycolysis kinetic model.

Lu, H., Li, F., Sánchez, B.J., *et al.* (2019). A consensus S. cerevisiae metabolic model Yeast8 and its ecosystem for comprehensively probing cellular metabolism. *Nature Communications* 10, 3586. yeast-GEM, SysBioChalmers.

Link, H., Fuhrer, T., Gerosa, L., Zamboni, N., & Sauer, U. (2015). Real-time metabolome profiling of the metabolic switch between starvation and growth. *Nature Methods* 12, 1091–1097. `https://www.nature.com/articles/nmeth.3584`.

Nardin, M., Wang, N., Elziny, S., *et al.* (2025). Reconstructing a physiological state space via chronic jugular microdialysis in freely moving mice. *bioRxiv*. `https://doi.org/10.64898/2025.12.08.692974`.